

9L. Access Methods

Introduction

"Access Methods" constitute the software abstraction layer—comprising science gateways, programmable APIs, and identity fabrics—that bridges the gap between complex Exascale infrastructure and diverse user communities. Transitioning from traditional command-line interfaces to these user-centric models is pivotal for the National Supercomputing Mission (NSM). This vertical aims to democratize access, transforming PARAM systems into service-oriented platforms that offer secure, seamless entry for AI researchers, MSMEs, and domain scientists while maintaining rigorous security across multi-center deployments.

Advanced Technologies ready to mature and commercialization

These technologies are at TRL 8/9, represented by existing commercial/open-source solutions, and are ready for immediate deployment in NSM 2.0.

- **Unified Web-Based Science Gateways**

Browser-based "single-pane-of-glass" interfaces allowing users to manage files, submit jobs, and visualize data without terminal knowledge.

International: Open OnDemand (NSF) - <https://www.openondemand.org/> , NICE EnginFrame (Commercial)(AWS - Discontinued since 25th Sept 2024) - <https://aws.amazon.com/blogs/hpc/discontinuation-of-nice-enginframe-effective-september-25th-2025/> , Altair Access - <https://altair.com/access/> .

National: NPSF TAS most commonly refers to the National PARAM Supercomputing Facility (NPSF) Technical Affiliation Scheme (TAS) , ParaDE(Parallel Dev. Env.) (C-DAC).

- **Interactive HPC & Virtual Desktop Infrastructure (VDI)**

Integration of notebook environments and remote visualization directly on compute nodes for real-time AI/ML exploration.

International: JupyterHub (Project Jupyter) - <https://jupyter.org/hub> , RStudio Server - <https://posit.co/products/open-source/rstudio-server/> , OpenShift (Red Hat) - <https://docs.redhat.com/en/products> .

National:

- **API-Driven HPC-as-a-Service (HPCaaS)**

Standardized RESTful APIs enabling external software, CI/CD pipelines, and cloud platforms to programmatically burst workloads to supercomputers.

International: FirecREST (CSCS) - <https://www.cscs.ch/services/products/firecrest> , Tapis Framework (UT Austin) - <https://tapis-project.org/> , AWS ParallelCluster - <https://aws.amazon.com/hpc/parallelcluster/> .

National:

- **Federated Identity and Access Management (IAM)**

SSO frameworks allowing seamless mobility of researchers across institutions using a single credential set.

International: CILogon - <https://www.cilogon.org/> , Keycloak - <https://www.keycloak.org/> , Globus Auth - <https://www.globus.org/globus-auth-service> , eduGAIN - <https://edugain.org/> .

National: Integration with Jan Parichay (NIC) and NSM-Grid Identity Fabric for unified access across IITs and C-DAC labs.

Futuristic Technologies on the Horizon

These technologies require R&D under NSM 2.0 for deployment in the NSM 3.0 timeframe (7–8 year horizon).

- **Intent-Based and AI-Mediated Access**

Natural Language Processing (NLP) interfaces where users declare intent (e.g., "Run climate model for Year 2050"), and AI agents autonomously handle resource allocation and script generation.

R&D Direction: Developing LLM-integrated shells and autonomous resource brokers.

- **Serverless HPC and Function-as-a-Service (FaaS)**

Event-driven models where users deploy lightweight code functions rather than full applications, utilizing ephemeral containers for micro-second latency.

R&D Direction: Adapting WebAssembly (Wasm) for high-performance interconnects and heterogeneous accelerators.

- **Zero-Trust and Quantum-Safe Access Fabrics**

Security architectures requiring continuous, context-aware authentication for every API call, secured by post-quantum cryptography.

R&D Direction: Integration of Quantum Key Distribution (QKD) protocols into standard SSH/HTTPS access layers.

- **National Federated Meta-Computing Grid**

A unified abstraction layer treating geographically distributed supercomputers as a single "Meta-System" with a unified namespace.

R&D Direction: Creation of meta-schedulers for dynamic workload routing between C-DAC centers based on energy efficiency and load balancing.